Correlation & Hypothesis Testing

Perform the hypothesis test using the given dataset.

Note: You have to use Marks.csv

Lab Task – 1 - **Hypothesis Test for Nominal variable**

Purpose: There is an assumption at HCT that the students who performed well on the project will score high marks in FWA. Is this assumption correct?
How To: Hypothesis Test using Correlation test - Pearson and Spearman's Rank Correlation

Answer the following questions:

A. Null Hypothesis: <<write here>>
B. Alternative Hypothesis: <<write here>>

C. Complete the Python program to accept or reject the hypothesis based on the p-value. The steps for programming and the sample code are provided to you.

D. Solution :(write below based on the output of your python program)

- Chi-Square Statistic: **<<write here>>**
- P-Value: **<<write here>>**
- Degrees of Freedom: **<<write here>>**

E. Conclusion :

- The p-value is much **<<write here>>**(lesser/greater) than the typical significance level of 0.05.
- This indicates that there is **<<write here>>**(significant/no significant) association between Final and Quiz.
- Thus, the null hypothesis is **<<write here>>**(rejected/accepted)

Answer:
A. Null Hypothesis: There is no significant association between Project marks and Final marks.
B. Alternative Hypothesis: There is a significant association between Project marks and Final marks.

a.	Check Project and Final are correlated visually – using scatter plot
	`# extract project and final samples and draw scatter plot`

```
# import the marks dataset
import pandas as pd
import matplotlib.pyplot as plt

# Load the data from the CSV file
marks = pd.read_csv("Marks.csv")
Project = marks[1:30]['Project']
Final = marks[1:30]['Final']

# Extract the columns for the x and y axes
x_values = Project
y_values = Final

# Create the scatter plot
plt.scatter(x_values, y_values)

# Add labels and a title
plt.xlabel("Project")
plt.ylabel("Final")
plt.title("Scatter Plot")

# Show the plot
plt.show()
```

Note: We are extracting only 30 records of **sample** from project and Final **population**. You can also use all the records without mentioning the number of rows.

b. Perform Spearman Rank Correlation to check project and final are correlated

A. **Null Hypothesis (H_0):** There is no significant correlation between Project marks and Final marks.
B. **Alternative Hypothesis (H_1):** There is a significant correlation between Project marks and Final marks.

```
#Spearman Rank Correlation
import pandas as pd
from scipy.stats import spearmanr

# Load the dataset
file_path = "Marks.csv"
df = pd.read_csv(file_path)

# Drop rows with NaN values for Project and Final
data = df[['Project', 'Final']].dropna()

# Perform Spearman's Correlation Test
r, p_value = spearmanr(data['Project'], data['Final'])
df_spearman = len(data) - 2

# Print the results
```

```
print("Spearman's Correlation Test Results:")
print(f"Spearman's Correlation Coefficient (r): {r:.4f}")
print(f"P-Value: {p_value:.4f}")
print(f"Degrees of Freedom: {df_spearman}")

# Interpretation
if p_value < 0.05:
    print("\nConclusion: Since the p-value is less than 0.05, we
reject the null hypothesis.")
    print("There is a statistically significant correlation
between Project and Final marks.")
else:
    print("\nConclusion: Since the p-value is greater than 0.05,
we accept the null hypothesis.")
    print("There is no statistically significant correlation
between Project and Final marks.")
```

```
Spearman's Correlation Test Results:
Spearman's Correlation Coefficient (r): 0.1700
P-Value: 0.4380
Degrees of Freedom: 21

Conclusion: Since the p-value is greater than 0.05, we accept the null hypothesis.
There is no statistically significant correlation between Project and Final marks.
```

E. Conclusion

The p-value is **greater** than the typical significance level of 0.05.

This indicates that there is **no significant** association between Project marks and Final marks.

Thus, the null hypothesis is **accepted**.

c. Perform Pearson Correlation to check project and final are correlated

A. Null Hypothesis: There is no significant association between Project marks and Final marks.

B. Alternative Hypothesis: There is a significant association between Project marks and Final marks.

```
#pearson correlation
from scipy.stats import pearsonr
import pandas as pd

# Load the dataset
df = pd.read_csv("Marks.csv")

# Extract Project and Final columns, dropping NaN values
project_scores = df['Project'].dropna()
final_scores = df['Final'].dropna()

# Ensure both series have equal length
min_length = min(len(project_scores), len(final_scores))
project_scores = project_scores[:min_length]
final_scores = final_scores[:min_length]
```

```
# Calculate Pearson correlation
r_value, p_value = pearsonr(project_scores, final_scores)
degrees_of_freedom = len(project_scores) - 2

# Print results
print(f"Pearson Correlation Coefficient (r): {r_value:.3f}")
print(f"P-Value: {p_value:.3f}")
print(f"Degrees of Freedom: {degrees_of_freedom}")

# Interpretation
if p_value < 0.05:
    print("\nConclusion: Since the p-value is less than 0.05, we reject the null hypothesis.")
    print("There is a statistically significant correlation between Project and Final marks.")
else:
    print("\nConclusion: Since the p-value is greater than 0.05, we accept the null hypothesis.")
    print("There is no statistically significant correlation between Project and Final marks.")
```

```
Pearson Correlation Coefficient (r): 0.253
P-Value: 0.244
Degrees of Freedom: 21

Conclusion: Since the p-value is greater than 0.05, we accept the null hypothesis.
There is no statistically significant correlation between Project and Final marks.
```

E. Conclusion

The p-value is **greater** than the typical significance level of 0.05.

This indicates that there is **no significant** association between Project marks and Final marks.

Thus, the null hypothesis is **accepted**.

d. Are other columns like Quiz, Practical correlate with Final using Pearson correlation?

```
marks[1:30].corr(method="pearson")
```

	Quiz	Practical	Project	Final
Quiz	1.000000	0.357823	0.389429	0.332885
Practical	0.357823	1.000000	0.400262	0.341331
Project	0.389429	0.400262	1.000000	0.251626
Final	0.332885	0.341331	0.251626	1.000000

Note: From the result, we can infer that Quiz-Practical(35%),Quiz-Project(38%), Quiz-Final(33%), Practical-Project(40%), Practical-Final(34%) are not correlated.

QUIZ, Practical,Project and Final are continuous Columns.

Lab Task – 2 - **Hypothesis Test for categorical variable**

Purpose: To check whether loan purpose is independent or dependent on loan status.

Assumption: Independent observation, size in each box of contingency table > 25

How To: Perform Correlation test -Chi-square Test

A. **Null Hypothesis:** Loan purpose is independent of loan status.
B. **Alternative Hypothesis:** Loan purpose is dependent on loan status.

a.	Importing the libraries
	`import pandas as pd`
b.	Importing the dataset
	`LoanData = pd.read_csv("Loan.csv")` `print(LoanData)`
c.	Derive Cross Tabulation report for loan status and purpose field

	`contingency_data = pd.crosstab(LoanData['purpose'], LoanData['loan_status'],margins = False)`
	`print(contingency_data)`
	loan_status Charged Off Fully Paid
	purpose
	credit_Card 14 42
	debt_consolidation 8 36
	Note: We assume that contingency table entry> 25 but some of the entries are less than 25. So, chi-square may be violated.

d.	Perform Chi- squared test for contigency table.

	`from scipy.stats import chi2_contingency`
	`r, p, dof, expected = chi2_contingency(contingency_data)`
	`print('r=%.3f, p=%.3f' % (r, p))`
	`if p > 0.05:`
	`  print('independent categories')`
	`else:`
	`  print('dependent categories')`
	
	`r=0.000, p=1.000`
	`independent categories`
	
	E. Conclusion
	
	The p-value is **greater** than the typical significance level of 0.05.
	This indicates that there is **no significant** association between loan purpose and loan status.
	
	Thus, the null hypothesis is **accepted**.

See Hypothesis report for Lab Task – 2 at the end of this exercise.

Lab Task – 3 **One Sample t-test (based on mean)**

HCT wants to assess the performance of FWA marks of the CIS students. HCT wants to know whether its sample is representative of the normal population. For a one-sample t-

test, there will only be one variable's data to be entered which is Final marks of CIS students.

Purpose: To find the average(mean) score of FWA by CIS students is same as average of all HCT students FWA marks.
Assumption: Normally distributed
How To: One sample T-test

Steps:

1. Find the average of final marks of all the students
2. Extract the final marks of CIS students
3. Check whether final marks are normally distributed using Shapiro-Wilkson method
4. Perform one sample T-test

Null hypothesis: The average score of FWA for CIS students is same as the average score of FWA for all HCT students.
Alternative hypothesis: The average score of FWA for CIS students is not same as the average score of FWA for all HCT students.

a.	Importing the libraries
	`import pandas as pd`
b.	Importing the FWA marks of all HCT Students

	`marks = pd.read_csv("Marks.csv")`
	`Final=marks['Final']`
	`print(Final)`

c.	Find the average(mean) of FWA for all HCT students

	`print(Final.mean())`
	`75.69565217391305`
	This result has to be applied in (f)

d.	Importing the FWA marks of CIS Students
	`Final_CIS=marks[marks['Division']=='CIS']` `Final_CIS['Final']`
e.	Perform whether FWA marks are normally distributed using shapiro method

	`from scipy.stats import shapiro`
	`r, p = shapiro(Final)`
	`print('r=%.2f, p=%.30f' % (r, p))`
	`if p > 0.05:`
	`  print('Normal distribution')`
	`else:`
	`  print('Not a normal distribution')`
	
	`r=0.94, p=0.202803328633308410644531250006`
	`Normal distribution`
	Note: Our assumption is true

f.	Conduct one sample T-test on FWA marks of CIS students
	`import scipy` `scipy.stats.ttest_1samp(Final_CIS['Final'],75)`

```
TtestResult(statistic=0.0, pvalue=1.0, df=2)
```

Note: As the p_value for the given problem is more than 0.05, we accept the null hypothesis and the alternative hypothesis is rejected.

75 is the average(mean) of FWA for all HCT students(which we derive from (c))
df = degree of freedom = no of samples – 1 [no of samples of CIS=3, so 3-1=2]

Hypothesis Testing Report

Subject: Performance of HCT students in the FWA exam

Background:

The HCT is considering the performance of FWA by their students. It wants to analyze whether there is any correlation exists between the project and FWA marks.

Hypotheses:

- Null Hypothesis (H0): The correlation exists between the Project and FWA marks
- Alternative Hypothesis (H1): The correlation does not exist between the Project and FWA marks

Test Statistic:

The test statistic is the sample proportion of 30 students who performed both Project and FWA.

P**-value**:

The correlation was calculated using both Pearson and Spearman methods and the p-value is calculated. The Pearson correlation was 0.25 and the Spearman correlation was 0.186. The p-value for the Pearson correlation was 0.25 and for Spearman correlation was 0.40.

Decision:

Since the p-value is greater than the significance level of 0.05 for both correlations, we reject the null hypothesis. This means that there is no correlation between the FWA marks and project marks.

Conclusion:

The results of this hypothesis test suggest that irrespective of scoring project marks high or low, the score of FWA does not have any impact. Therefore, the university should consider revising the FWA question patterns.

Recommendation:

I suggest that the HCT implement case study questions that enhance students' cognitive and analytic abilities. Furthermore, it will aid students in transferring

course-learned knowledge to industrial needs. In addition, it will assist in removing the bias that only a few students actively participated in the team, while others did not in the group project.